4. Pollen tube was discovered by
   a) J.G.Kolreuter b) G.B.Amici
   c) E.Strasburger d) E.Hanning

5. Size of pollen grain in *Myosotis*
   a) 10 micrometer b) 20 micrometer
   c) 200 micrometer d) 2000 micrometer

6. First cell of male gametophyte in angiosperm is
   a) Microspore b) megaspore
   c) Nucleus d) Primary Endosperm Nucleus

7. Match the following

I) External fertilization	i) pollen grain
II) Androecium	ii)anther wall
III) Male gametophyte	iii)algae
IV) Primary parietal layer	iv)stamens

   a)I-iv;II-i;III-ii;IV-iii
   b)I-iii;II-iv;III-i;IV-ii
   c)I-iii;II-iv;III-ii,IV-i
   d)I-iii;II-i;III-iv;IV-ii

8. Arrange the layers of anther wall from locus to periphery
   a) Epidermis,middle layers, tapetum, endothecium
   b) Tapetum, middle layers, epidermis, endothecium
   c) Endothecium, epidermis, middle layers, tapetum
   d) Tapetum, middle layers endothecium epidermis

9. Identify the incorrect pair
   a) sporopollenin - exine of pollen grain
   b) tapetum – nutritive tissue for developing microspores
   c) Nucellus – nutritive tissue for developing embryo
   d) obturator – directs the pollen tube into micropyle

10. Assertion : Sporopollenin preserves pollen in fossil deposits
    Reason : Sporopollenin is resistant to physical and biological decomposition
    a) assertion is true; reason is false
    b) assertion is false; reason is true
    c) Both Assertion and reason are not true
    d) Both Assertion and reason are true.

11. Choose the correct statement(s) about tenuinucellate ovule
    a) Sporogenous cell is hypodermal
    b) Ovules have fairly large nucellus
    c) sporogenous cell is epidermal
    d) ovules have single layer of nucellus tissue

12. The correct order of haploid, diploid and triploid structure in fertilized embryo sac is
    a) synergid, zygote and PEN
    b) synergid, antipodal and polar nuclei
    c) antipodal, synergid and PEN
    d) synergid, polar nuclei and zygote

13. Which of the following represent megagametophyte
    a) Ovule b)Embryo sac
    c)Nucellus d)Endosperm

14. In *Haplopappus gracilis*, number of chromosomes in cells of nucellus is 4. What will be the chromosome number in Primary endosperm cell?
    a)8 b)12 c)6 d)2

15. Transmitting tissue is found in
    a) Micropylar region of ovule
    b) Pollen tube wall
    c) Stylar region of gynoecium
    d) Integument

16. The scar left by funiculus in the seed is
    a)tegmen b)radicle
    c)epicotyl d)hilum

17. A Plant called X possesses small flower with reduced perianth and versatile anther. The probable agent for pollination would be
    a)water b)air
    c)butterflies d)beetles

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